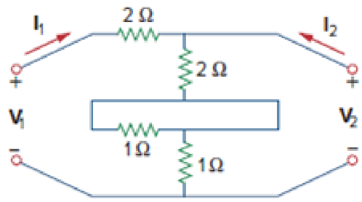
Chapter 19, Problem 65P

Problem

What is the *y* parameter presentation of the circuit in Fig. 19.112?

Figure 19.112



Step-by-step solution

Step 1 of 5

Consider the following circuit:

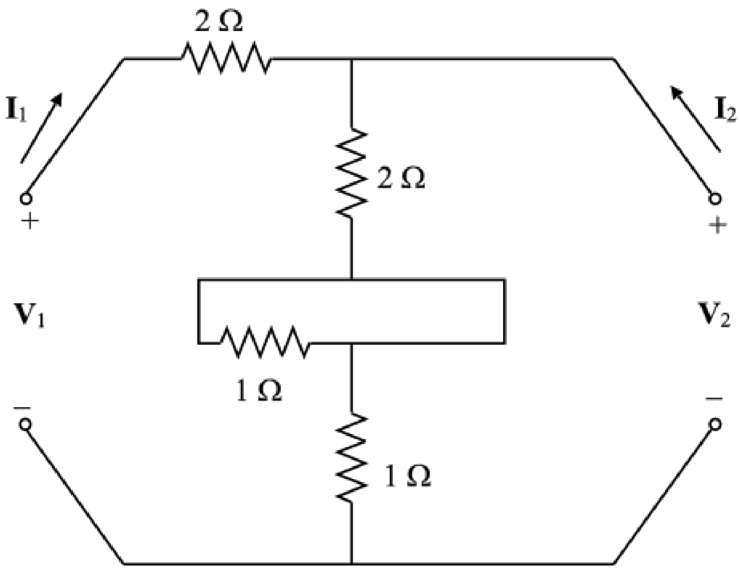


Figure 1

Step 2 of 5

Determine *z* parameters first and then convert these into *y* parameters.

Define *z* parameters as follows.

$$\mathbf{V}_1 = \mathbf{z}_{11}\mathbf{I}_1 + \mathbf{z}_{12}\mathbf{I}_2$$
$$\mathbf{V}_2 = \mathbf{z}_{21}\mathbf{I}_1 + \mathbf{z}_{22}\mathbf{I}_2$$

$$\mathbf{z}_{11} = \left.\frac{\mathbf{V}_1}{\mathbf{I}_1}\right|_{\mathbf{I}_2=0}$$

$$\mathbf{z}_{12} = \left.\frac{\mathbf{V}_1}{\mathbf{I}_2}\right|_{\mathbf{I}_1=0}$$

$$\mathbf{z}_{21} = \left.\frac{\mathbf{V}_2}{\mathbf{I}_1}\right|_{\mathbf{I}_2=0}$$

$$\mathbf{z}_{22} = \left.\frac{\mathbf{V}_2}{\mathbf{I}_2}\right|_{\mathbf{I}_1=0}$$

To determine $\mathbf{z}_{11}$ and $\mathbf{z}_{21}$, apply voltage source $\mathbf{V}_1$ and leave the output port open.

Apply Kirchhoff's voltage law to the input loop.

$$\mathbf{V}_1 = (2+2+1)\mathbf{I}_1$$
$$\mathbf{V}_1 = 5\mathbf{I}_1$$
$$\left.\frac{\mathbf{V}_1}{\mathbf{I}_1}\right|_{\mathbf{I}_2=0} = 5$$

As there are two paths for the current after $2\ \Omega$ resistor in the parallel branch, the current takes the path with zero resistance. Thus, $1\ \Omega$ does not have any current through it.

Step 3 of 5

Apply Kirchhoff's current law to the top node.

$$\frac{\mathbf{V}_1 - \mathbf{V}_2}{2} = \frac{\mathbf{V}_2}{2+1}$$
$$3\mathbf{V}_1 - 3\mathbf{V}_2 = 2\mathbf{V}_2$$
$$3\mathbf{V}_1 - 5\mathbf{V}_2 = 0$$

Substitute $5\mathbf{I}_1$ for $\mathbf{V}_1$.

$$3(5\mathbf{I}_1) - 5\mathbf{V}_2 = 0$$
$$15\mathbf{I}_1 = 5\mathbf{V}_2$$
$$\left.\frac{\mathbf{V}_2}{\mathbf{I}_1}\right|_{\mathbf{I}_2=0} = 3$$

Step 4 of 5

To determine $\mathbf{z}_{12}$ and $\mathbf{z}_{22}$, apply voltage source $\mathbf{V}_2$ to the output port and leave the input port open.

Apply Kirchhoff's voltage law to the output loop.

$$\mathbf{V}_2 = (2+1)\mathbf{I}_2$$
$$\mathbf{V}_2 = 3\mathbf{I}_2$$
$$\left.\frac{\mathbf{V}_2}{\mathbf{I}_2}\right|_{\mathbf{I}_1=0} = 3$$

And

$$\mathbf{V}_1 = (2+1)\mathbf{I}_2$$
$$\left.\frac{\mathbf{V}_1}{\mathbf{I}_2}\right|_{\mathbf{I}_1=0} = 3$$

The *z* parameters of the network are $\begin{bmatrix} 5 & 3 \\ 3 & 3 \end{bmatrix}$.

Write the conversion formulae to find *y*-parameters.

$$y_{11} = \frac{z_{22}}{z_{11}z_{22} - z_{12}z_{21}}$$
$$y_{12} = -\frac{z_{12}}{z_{11}z_{22} - z_{12}z_{21}}$$
$$y_{21} = -\frac{z_{21}}{z_{11}z_{22} - z_{12}z_{21}}$$
$$y_{22} = \frac{z_{11}}{z_{11}z_{22} - z_{12}z_{21}}$$

Substitute *z* parameter values.

$$\begin{aligned} y_{11} &= \frac{z_{22}}{z_{11}z_{22} - z_{12}z_{21}} \\ &= \frac{3}{(5)(3)-(3)(3)} \\ &= \frac{3}{15-9} \\ &= \frac{3}{6} \end{aligned}$$
$$y_{11} = \frac{1}{2}$$

$$\begin{aligned} y_{12} &= -\frac{z_{12}}{z_{11}z_{22} - z_{12}z_{21}} \\ &= -\frac{3}{(5)(3)-(3)(3)} \\ &= -\frac{3}{15-9} \\ &= -\frac{3}{6} \end{aligned}$$
$$y_{12} = -\frac{1}{2}$$

Step 5 of 5

$$
\begin{aligned}
y_{21} &= -\frac{z_{21}}{z_{11}z_{22} - z_{12}z_{21}} \\
&= -\frac{3}{(5)(3) - (3)(3)} \\
&= -\frac{3}{15 - 9} \\
&= -\frac{3}{6} \\
y_{21} &= -\frac{1}{2}
\end{aligned}
$$

$$
\begin{aligned}
y_{22} &= \frac{z_{11}}{z_{11}z_{22} - z_{12}z_{21}} \\
&= \frac{5}{(5)(3) - (3)(3)} \\
&= \frac{5}{15 - 9} \\
&= \frac{5}{6}
\end{aligned}
$$

Thus, the y parameters of the network are $\boxed{\begin{bmatrix} \frac{1}{2} & -\frac{1}{2} \\ -\frac{1}{2} & \frac{5}{6} \end{bmatrix}}$.